Definition/Introduction

General anesthesia is a medically-induced loss of consciousness with concurrent loss of protective reflexes due to anesthetic agents. Various medications may be prescribed to induce unconsciousness, amnesia, analgesia, skeletal muscle relaxation, and the loss of autonomic system reflexes.[1] During this state, the patient is unarousable to verbal, tactile, and painful stimuli. Upper airway obstruction during general anesthesia usually necessitates the insertion of a laryngeal mask airway or endotracheal tube to preserve airway patency. Likewise, the patient's spontaneous ventilation is often inadequate, requiring partial or full mechanical support with positive pressure ventilation. The patient's cardiovascular function may also become impaired.

In the past, when physical examination offered the only clue as to a patient's depth of anesthesia, anesthetic overdose by an inexperienced anesthetist easily occurred. It was not until the 20th century that the anesthesia community developed a truly systematic approach to monitoring. In 1937, Dr. Arthur Guedel created one of the first safety systems in anesthesiology, with a chart that explained the stages of anesthesia with an increasing depth ranging from stages 1 to 4.[2] Despite newer anesthetic medications and delivery techniques that have led to faster onset and recovery from general anesthesia (and in some cases circumventing certain stages entirely), Guedel's classification is still used.

Stages of Anesthesia Based on Guedel's Classification

- Stage 1 - Analgesia or Disorientation: This stage can be initiated in a preoperative anesthesiology holding area, where the patient is given medication and may begin to feel its effects but has not yet become unconscious. This stage is usually described as the "induction stage." Patients are sedated but conversational. Breathing is slow and regular. At this stage, the patient progresses from analgesia free of amnesia to analgesia with concurrent amnesia.[3] This stage comes to an end with the loss of consciousness.

- Stage 2 - Excitement or Delirium: This stage is marked by features such as disinhibition, delirium, uncontrolled movements, loss of eyelash reflex, hypertension, and tachycardia. Airway reflexes remain intact during this phase and are often hypersensitive to stimulation. Airway manipulation during this stage of anesthesia should be avoided, including both the placement and removal of endotracheal tubes and deep suctioning maneuvers. There is a higher risk of laryngospasm (involuntary tonic closure of vocal cords) at this stage, which may be aggravated by any airway manipulation. Consequently, the combination of spastic movements, vomiting, and rapid, irregular respirations can compromise the patient's airway.[4] Fast-acting agents help reduce the time spent in stage 2 as much as possible and facilitate entry to stage 3.

- Stage 3 – Surgical Anesthesia: This is the targeted anesthetic level for procedures requiring general anesthesia. Ceased eye movements and respiratory depression are the hallmarks of this stage. Airway manipulation is safe at this level. There are four "planes" described for this stage.[5] During plane 1, there is still regular spontaneous breathing, constricted pupils, and central gaze. However, eyelid, conjunctival, and swallow reflexes usually disappear in this plane. During plane 2, there are intermittent cessations of respiration along with the loss of corneal and laryngeal reflexes. Halted ocular movements and increased lacrimation may also occur. Plane 3 is marked by complete relaxation of the intercostal and abdominal muscles and loss of the pupillary light reflex. This plane is referred to as "true surgical anesthesia" because it is ideal for most surgeries. Finally, Plane 4 is marked by irregular respiration, paradoxical rib cage movement, and full diaphragm paralysis resulting in apnea.[4][5]

- Stage 4 - Overdose: This stage occurs when too much anesthetic agent is given relative to the amount of surgical stimulation, which results in worsening of an already severe brain or medullary depression. This stage begins with respiratory cessation and ends with potential death. Skeletal muscles are flaccid, and pupils are fixed and dilated at this stage.[4][6] Blood pressure is typically significantly lower than normal, with weak and thready pulses due to the suppression of the cardiac pump and vasodilation in the peripheral bloodstream. Without cardiovascular and respiratory support, this stage is lethal. Hence, the anesthetist's goal is to transition the patient as soon as possible to stage 3 of anesthesia and keep them there for the duration of the operation.

Issues of Concern

General anesthesia induces physiological responses, potentially resulting in morbidity and mortality when emergency situations are not properly addressed. Therefore, it is regarded as a high-risk activity where the benefits of surgery must outweigh the potential harms. Mortality directly related to anesthetic management is rare but may result from pulmonary aspiration of gastric contents, asphyxiation, or anaphylaxis.[7] These adverse events may result from anesthesia-related equipment failure or, more often, from human error. However, numerous investigators and professional organizations have stated that mortality rates attributable to anesthesia have decreased over the last two decades.[8] This decrease is reportedly due to advances in safety, including enhanced detection and monitoring methods and new technology, modernization, and extensive adoption of practice guidelines and other quality improvement measures to reduce errors. Anesthesia today is generally considered safe and effective, especially when practiced by an experienced and well-prepared anesthesia provider.[8][9]

Clinical Significance

Guedel's classification for the stages of general anesthesia was initially established to deliver diethyl ether, the single available volatile anesthetic at the time. While Gaudel's patients were usually pre-medicated with sedative agents such as morphine and atropine, ether was the drug of choice for induction.[10] It offered analgesia, amnesia, and relaxation of muscles. However, ether was phased out in the United States by the 1980s and replaced with the current fluorinated hydrocarbon anesthetics. Today, the "balanced anesthesia" approach uses several types of medications for induction (such as intravenous anesthetics, analgesics, neuromuscular blockers, and benzodiazepines), which can disguise the characteristic clinical markers of each defined anesthesia stage. These agents also have a higher safety profile than diethyl ether. Lastly, advances in awareness, breathing, and circulation monitoring due to technology have greatly augmented the clinical data obtained from the patient's physical examination. Therefore, some anesthesiologists view Guedel's work as antiquated. Yet others still employ his classification to describe developments in general anesthesia and clinical practice for inhalation inductions across various surgical procedures.[11][12]

Nursing, Allied Health, and Interprofessional Team Interventions

Anesthesia is thought to be best practiced via an interdisciplinary approach that includes anesthesiologists and certified registered nurse anesthetists, nurses and other operating room staff, recovery room nurses, and anesthesia technicians to ensure patient safety. Because there is no agent capable of instantly reversing the effects of inhaled anesthetics, close monitoring of the patient is necessary during anesthesia. A responsible anesthesiologist, anesthetist, or nurse should attentively monitor vital signs during induction and maintenance periods to confirm that the patient is appropriately sedated without signs of instability.[13] Standard guidelines and regulations for monitoring patients during anesthesia are required in every hospital operating room, outpatient surgical or procedure center, and office-based setting. The benefits of collaboration in healthcare positively correlate with increased patient satisfaction, improved patient outcomes, increased staff satisfaction, and reduced hospital costs.